

Problem Set 5. Linear Regression

Quantitative Methods for Humanities

1. University Life and Final Grade.

A university wants to study whether weekly hours spent in student associations are related to students' final grade in an Economics course. For a sample of students, the following simple linear regression is estimated:

$$\widehat{\text{grade}} = 5.8 + 0.22 \text{association_hours}.$$

Grades are measured from 0 to 10, and **association_hours** measures hours per week.

- (a) Interpret the intercept and the slope coefficient in this context.
- (b) A student who spends 18 hours per week in student associations obtains a final grade of 9.44. Does the linear model under or overestimate the student's performance in the final exam?
- (c) The observed values of **association_hours** in the sample range from 0 to 12. Was the prediction for the student in the previous part reliable? Explain.

2. Predicting Support for Rent Control Policies.

A political scientist studies whether citizens who spend a larger share of their income on rent are more likely to support rent control policies. The variable **rent_burden** measures the percentage of monthly income spent on rent. The variable **urban** is a dummy equal to 1 if the respondent lives in a large city and 0 otherwise.

The researcher estimates the following model using survey data:

$$\widehat{\text{support}} = 22 + 0.85 \text{rent_burden} + 9 \text{urban}.$$

Support for rent control is measured on a 0–100 scale. The sample used to estimate the model contains values of **rent_burden** between 10 and 55.

- (a) Interpret the coefficient on **rent_burden**.
- (b) Interpret the coefficient on **urban**.
- (c) What is the predicted difference between the a non-urban and an urban respondent who spend the same percentage of their income on rent?
- (d) A new respondent spends 65% of their income on rent. Is this prediction within range or outside range? Why does this matter?

- (e) The researcher worries that the relationship between rent burden and support for rent control is not linear. Write a transformed linear model that allows rent burden to have a nonlinear effect while remaining a linear regression model in the parameters.

3. Civic Education Workshops: Regression and Difference in Means.

A political science department runs a randomized experiment to study whether civic education workshops increase citizens' political knowledge. Some citizens are randomly assigned to attend a short workshop about local institutions, while others are not invited. Let Y be the number of correct answers in a political knowledge quiz for a citizen, and let $T = 1$ if the citizen was assigned to the workshop and $T = 0$ otherwise.

The data are summarized as follows:

Group	Number of citizens	Mean of Y	Mean baseline interest in politics
Control, $T = 0$	120	5.2	3.0
Treatment, $T = 1$	125	6.1	3.1

- (a) Consider the regression $Y = \beta_0 + \beta_1 T + u$. What are the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$?
 (b) Interpret $\hat{\beta}_1$ as a difference in means. Under random assignment, what is its causal interpretation?
 (c) Why is random assignment important for this causal interpretation?
 (d) A second researcher estimates

$$Y = \beta_0 + \beta_1 T + \beta_2 \text{interest} + \beta_3 (T \cdot \text{interest}) + u.$$

Interpret β_3 .

- (e) According to the interaction model, what is the treatment effect for a citizen with interest = 4? Give the formula.

4. Scholarships and a Regression Discontinuity Design.

A regional government awards a university scholarship to students who score at least 80 points on an entrance exam. Let X be the exam score, let $T = 1$ if $X \geq 80$, and let Y be first-year university enrollment, measured as 1 if the student enrolls and 0 otherwise.

The researcher estimates the following regression:

$$Y = \beta_0 + \beta_1 X + \beta_2 T + \beta_3 (T \cdot X) + u.$$

The estimated coefficients are:

$$\hat{\beta}_0 = -0.55, \quad \hat{\beta}_1 = 0.015, \quad \hat{\beta}_2 = 0.30, \quad \hat{\beta}_3 = -0.001.$$

- (a) Compute the estimated jump at the cutoff $c = 80$.
 (b) State the key assumption that makes this jump a valid estimate of the average treatment effect.
 (c) Explain why this RDD estimate may have strong internal validity near the cutoff but weaker external validity for students with scores far from 80.